Project Design Phase
Solution Architecture

Date	7 February 2026
Team ID	LTVIP2026TMIDS86321
Project Name	**IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro**
Maximum Marks	4 Marks

Solution Architecture:

- Solution architecture describes the overall structure of the IntelliSQL system and how different components interact to solve the problem of complex SQL querying. It explains how natural language queries are processed, converted into SQL queries, executed on the database, and how the results are displayed to the user. The main goals of the solution architecture are:
- • Identify the most suitable technological approach to simplify database querying for users.
- • Describe the system structure, components, and data flow involved in processing natural language queries.
- • Define the system modules such as the Streamlit interface, query processing module, SQL generator, and database.
- • Provide clear specifications for how the system processes user queries, interacts with the database, and delivers results to the user.
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Example - Solution Architecture Diagram:

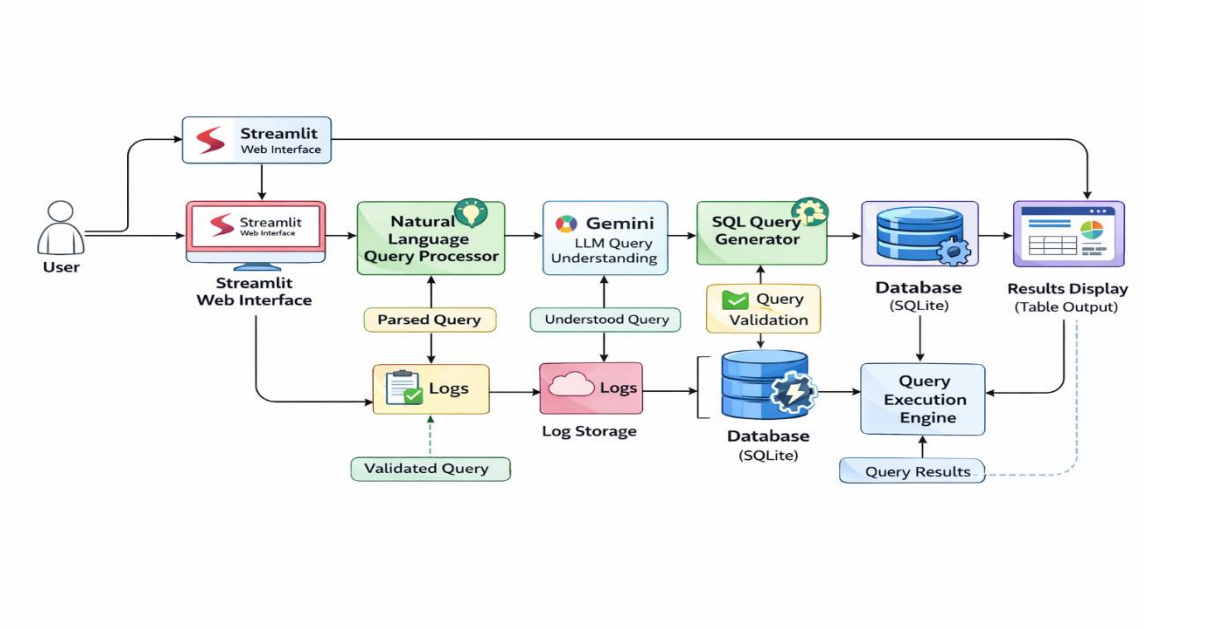

Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/